ECON 609: Heterogeneous Agent Models

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Lecture # 10

Introduction

- Often, we want to deal with models with heterogeneous agents.
- Examples:
  - Heterogeneity in age: OLG, life-cycle models
  - Heterogeneity in policies: progressive marginal tax rates
- These models are suitable to study issues regarding inequality.
  - For instance, endogenous consumption and wealth distribution are generated.
  - This enables meaningful policy experiments that affect distributions.

Models without Aggregate Uncertainty I

- Continuum of measure 1 of individuals, each facing an income fluctuation problem.
- Labor income: $w_t y_t$
- Labor endowment process $\{y_t\}_{t=0}^{\infty}$, $y_t \in Y = \{y_1, \ldots, y_N\}$.
- Labor endowment process follows stationary Markov chain with transitions $\pi(y'|y)$.
- Π denotes stationary distribution associated with π.

Models without Aggregate Uncertainty II

- Total labor endowment in the economy at each point of time

$$\bar{L} = \sum_y y\Pi(y)$$

- Probability of event history y^t, given initial event y_0

$$\pi_t(y^t|y_0) = \pi(y_t|y_{t-1}) \times \ldots \times \pi(y_1|y_0)$$

- Note that Markov structure is used.

- Substantial idiosyncratic uncertainty, but no aggregate uncertainty.

Models without Aggregate Uncertainty II

- Preferences
$$u(c) = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t)$$
- Budget constraint
$$c_t + a_{t+1} = w_t y_t + (1 + r_t) a_t$$
- Borrowing constraint
$$a_{t+1} \geq 0$$
- Initial conditions of agent (a_0, y_0) with initial population measure $\Phi_0(a_0, y_0)$.
- Allocation: $\{c_t(a_0, y^t), a_{t+1}(a_0, y^t)\}$.

Models with Aggregate Uncertainty IV

- Technology: $Y_t = F(K_t, L_t)$.
- Capital depreciates at rate $0 < \delta < 1$.
- Aggregate resource constraint

$$C_t + K_{t+1} - (1-\delta)K_t = F(K_t, L_t)$$

- The only asset in this economy is the physical capital stock. Incomplete market.

Recursive Equilibrium

- Individual state (a, y).
- Aggregate state variable $\Phi(a, y)$.
- $A = [0, \infty)$: set of possible asset holdings.
- Y: set of possible labor endowment realizations.

Household Problem in Recursive Formulation

$$v(a, y; \Phi) = \max_{c \geq 0, a' \geq 0} u(c) + \beta \sum_{y' \in Y} \pi(y'|y) v(a', y'; \Phi')$$

$$s.t.\ c + a' = w(\Phi)y + (1 + r(\Phi))a$$

$$\Phi' = H(\Phi)$$

- Function $H : \mathcal{M} \to \mathcal{M}$ is called the aggregate law of motion.

Definition of RCE

- A RCE is value function $v : Z \times \mathcal{M} \to \mathbb{R}$, policy functions for the household $a' : Z \times \mathcal{M} \to \mathbb{R}$ and $c : Z \times \mathcal{M} \to \mathbb{R}$, policy functions for the firm $K : \mathcal{M} \to \mathbb{R}$ and $L : \mathcal{M} \to \mathbb{R}$, pricing functions $r : \mathcal{M} \to \mathbb{R}$ and $w : \mathcal{M} \to \mathbb{R}$ and law of motion $H : \mathcal{M} \to \mathcal{M}$ s.t.

(1) v, a', c are measurable with respect to $\mathcal{B}(Z)$, v satisfies Bellman equation and a', c are policy functions given r and w.

(2) K, L satisfy

$$r(\Phi) = F_K(K(\Phi), L(\Phi)) - \delta$$
$$w(\Phi) = F_L(K(\Phi), L(\Phi))$$

Definition of RCE (cont.)

(3) For all $\Phi \in \mathcal{M}$,

$$K'(\Phi) = K(H(\Phi)) = \int a'(a, y; \Phi) d\Phi$$

$$L(\Phi) = \int y d\Phi$$

$$\int c(a, y; \Phi) d\Phi + \int a'(a, y; \Phi) d\Phi = F(K(\Phi), L(\Phi)) + (1 - \delta) K(\Phi)$$

(4) Aggregate law of motion H is generated by π and a'.

Transition Functions

- Define transition function $Q_\Phi : Z \times \mathcal{B}(Z) \to [0,1]$ by

$$Q_\Phi((a,y),(\mathcal{A},\mathcal{Y})) = \sum_{y' \in \mathcal{Y}} \begin{cases} \pi(y'|y) \text{ if } a'(a,y;\Phi) \in \mathcal{A} \\ 0 \qquad\qquad \text{otherwise} \end{cases}$$

  for all $(a,y) \in Z$ and all $(\mathcal{A},\mathcal{Y}) \in \mathcal{B}(Z)$.

- $Q_\Phi((a,y),(\mathcal{A},\mathcal{Y}))$ is the probability that an agent with current assets a and income y ends up with assets $a' \in \mathcal{A}$ and income $y' \in \mathcal{Y}$ tomorrow.

- Hence

$$\Phi'(\mathcal{A},\mathcal{Y}) = \int Q_\Phi((a,y),(\mathcal{A},\mathcal{Y}))\Phi(da \times dy)$$

Definition of Stationary RCE

- A stationary RCE is value function $v : Z \to \mathbb{R}$, policy functions for the household $a' : Z \to \mathbb{R}$ and $c : Z \to \mathbb{R}$, policies for the firm K, L, prices r, w and a measure $\Phi \in \mathcal{M}$ s.t.

(1) v, a', c are measurable with respect to $\mathcal{B}(Z)$, v satisfies the household's Bellman equation and a', c are the associated policy functions given r, w.

(2) K, L satisfy

$$r = F_K(K, L) - \delta$$
$$w = F_L(K, L)$$

Definition of Stationary RCE (cont.)

(3) The followings are satisfied:

$$
\begin{aligned}
K &= \int a'(a,y)d\Phi \\
L &= \int y d\Phi \\
\int c(a,y)d\Phi &+ \int a'(a,y)d\Phi = F(K,L) + (1-\delta)K
\end{aligned}
$$

(4) For all $(\mathcal{A},\mathcal{Y}) \in \mathcal{B}(Z)$,

$$\Phi(\mathcal{A},\mathcal{Y}) = \int Q((a,y),(\mathcal{A},\mathcal{Y}))d\Phi$$

where Q is transition function induced by π and a'.

Example: Discrete State Space

- Suppose $A = \{a_1, \ldots, a_M\}$. Then Φ is $M \times N$ column vector and $Q = (q_{ij,kl})$ is $(M \times N) \times (M \times N)$ matrix with

$$q_{ij,kl} = Pr((a', y') = (a_k, y_l)|(a, y) = (a_i, y_j))$$

- Stationary measure Φ satisfies $\Phi = Q^T \Phi$.
- Φ is eigenvector associated with an eigenvalue $\lambda = 1$ of Q^T.
- Q^T is a stochastic matrix and thus has at least one unit eigenvalue.

Some Comments

- Asset market clearing condition

$$K = K(r) = \int a'(a, y) d\Phi \equiv Ea(r)$$

- Labor market equilibrium $L = \bar{L}$ and $\bar{L}$ is exogenously given.
- Capital demand of firm $K(r)$ is defined implicitly as

$$r = F_K(K(r), \bar{L}) - \delta$$

Computation

- Fix an $r \in (-\delta, \frac{1}{\beta} - 1)$.
- For a fixed r, solve household's recursive problem. This yields a value function v_r and decision rules a'_r, c_r.
- The policy function a'_r and π induce Markov transition function Q_r.
- Compute the unique stationary measure Φ_r associated with this transition function.
- Compute excess demand for capital $d(r) = K(r) - Ea(r)$. If zero, stop, if not, adjust r.

Qualitative Results

- This model: $r^* < \frac{1}{\beta} - 1$.
- Overaccumulation of capital and oversaving (because of precautionary savings motive)
- Question: How big a difference does it make?

Calibration I

- CRRA with values $\sigma \in \{1, 3, 5\}$.
- $\beta = 0.96$.
- Cobb-Douglas production function with $\alpha = 0.36$.
- $\delta = 8\%$.
- Earnings profile:

$$\log(y_{t+1}) = \theta \log(y_t) + \sigma_\epsilon (1 - \theta^2)^{\frac{1}{2}} \epsilon_{t+1}.$$

- Consider $\theta \in \{0, 0.3, 0.6, 0.9\}$ and $\sigma_\epsilon \in \{0.2, 0.4\}$.

Calibration II

- Discretize, using Tauchen's method (Set $N = 7$.)
  - Since $\log(y_t) \in (-\infty, \infty)$, subdivide in intervals
$$(-\infty, -\frac{5}{2}\sigma_\epsilon), [-\frac{5}{2}\sigma_\epsilon, -\frac{3}{2}\sigma_\epsilon), \ldots, [\frac{3}{2}\sigma_\epsilon, \frac{5}{2}\sigma_\epsilon), [\frac{5}{2}\sigma_\epsilon, \infty)$$
  - State space for log-income: "midpoints"
$$Y^{\log} = \{-3\sigma_\epsilon, -2\sigma_\epsilon, -\sigma_\epsilon, 0, \sigma_\epsilon, 2\sigma_\epsilon, 3\sigma_\epsilon\}$$
  - Matrix π: fix $s_i = \log(y) \in Y^{\log}$ today and the conditional probability of $s_j = \log(y') \in Y^{\log}$ tomorrow is
$$\pi(\log(y') = s_j | \log(y) = s_i) = \int_{I_j} \frac{e^{-\frac{(x-\theta s_i)^2}{2\sigma_y}}}{(2\pi)^{0.5}\sigma_y} dx$$
where $\sigma_y = \sigma_\epsilon(1-\theta^2)^{\frac{1}{2}}$.

Calibration III

- Find the stationary distribution of π by solving

$$\Pi = \pi^T \Pi$$

- Take $\widetilde{Y} = e^{Y^{\log}}$

$$\widetilde{Y} = \{e^{-3\sigma_\epsilon}, e^{-2\sigma_\epsilon}, e^{-\sigma_\epsilon}, 1, e^{\sigma_\epsilon}, e^{2\sigma_\epsilon}, e^{3\sigma_\epsilon}\}$$

- Compute average labor endowment $\bar{y} = sum_{y \in \widetilde{Y}} y \Pi(y)$.
- Normalize all states by $\bar{y}$

$$Y = \{\frac{e^{-3\sigma_\epsilon}}{\bar{y}}, \frac{e^{-2\sigma_\epsilon}}{\bar{y}}, \frac{e^{-\sigma_\epsilon}}{\bar{y}}, \frac{1}{\bar{y}}, \frac{e^{\sigma_\epsilon}}{\bar{y}}, \frac{e^{2\sigma_\epsilon}}{\bar{y}}, \frac{e^{3\sigma_\epsilon}}{\bar{y}}\}$$

- Then $\sum_{y \in Y} y \Pi(y) = 1$.

Results

- With Cobb-Douglas production function and $\bar{L} = 1$, we have

$$Y = K^{\alpha}, \quad r + \delta = \alpha K^{\alpha - 1}$$

- What happens to capital with
  (1) an increase in θ keeping σ and σ_ϵ fixed?
  (2) an increase in σ keeping θ and σ_ϵ fixed?
  (3) an increase in σ_ϵ keeping σ and θ fixed?

Illustration

TABLE II

A. Net return to capital in %/aggregate saving rate in % ($\sigma = 0.2$)

$\rho\backslash\mu$	1	3	5
0	4.1666/23.67	4.1456/23.71	4.0858/23.83
0.3	4.1365/23.73	4.0432/23.91	3.9054/24.19
0.6	4.0912/23.82	3.8767/24.25	3.5857/24.86
0.9	3.9305/24.14	3.2903/25.51	2.5260/27.36

B. Net return to capital in %/aggregate saving rate in % ($\sigma = 0.4$)

$\rho\backslash\mu$	1	3	5
0	4.0649/23.87	3.7816/24.44	3.4177/25.22
0.3	3.9554/24.09	3.4188/25.22	2.8032/26.66
0.6	3.7567/24.50	2.7835/26.71	1.8070/29.37
0.9	3.3054/25.47	1.2894/31.00	−0.3456/37.63

Models with Aggregate Uncertainty

- Why complicate the model? Want to talk about economic fluctuations and its interaction with idiosyncratic uncertainty.
- But now we have to characterize and compute entire recursive equilibrium.
- Aggregate uncertainty and distribution as state variables.
- Distribution is infinite-dimensional object.

The Model I

- Aggregate production function:

$$Y_t = s_t F(K_t, L_t)$$

- Let $s_t \in \{s_b, s_g\} = S$ with $s_b < s_g$ and conditional probabilities $\pi(s'|s)$.

- Idiosyncratic labor productivity $y_t \in \{y_u, y_e\} = Y$ where $y_u < y_e$, and y_u stands for the agent being unemployed and y_e for the agent being employed.

- y_t is correlated with s_t: probability of being unemployed is higher during recessions than during expansions.

The Model II

- Probability of individual productivity y' and aggregate state s' tomorrow, conditional on states y and s today:

$$\pi(y', s'|y, s) \geq 0$$

- Law of large numbers: idiosyncratic uncertainty averages out and only aggregate uncertainty determines $\Pi_s(y)$.

- Consistency requires:

$$\sum_{y' \in Y} \pi(y', s'|y, s) = \pi(s'|s) \quad \forall y \in Y, \forall s, s' \in S$$

$$\Pi_{s'}(y') = \sum_{y \in Y} \frac{\pi(y', s'|y, s)}{\pi(s'|s)} \Pi_s(y) \quad \forall s, s' \in S$$

Recursive Formulation

- Individual state variables (a, y).
- Aggregate state variables (s, Φ).
- Recursive formulation:

$$v(a, y, s, \Phi) = \max_{c, a' \geq 0} \{U(c) + \beta \sum_{y' \in Y} \sum_{s' \in S} \pi(y', s' | y, s) v(a', y', s', \Phi')\}$$

$$s.t.\ c + a' = w(s, \Phi) y + (1 + r(s, \Phi)) a$$

$$\Phi' = H(s, \Phi, s')$$

Definition of RCE

- A RCE is value function $v : Z \times S \times \mathcal{M} \to \mathbb{R}$, policy functions for the household $a' : Z \times S \times \mathcal{M} \to \mathbb{R}$ and $c : Z \times S \times \mathcal{M} \to \mathbb{R}$, policy functions for the firm $K : S \times \mathcal{M} \to \mathbb{R}$ and $L : S \times \mathcal{M} \to \mathbb{R}$, pricing functions $r : S \times \mathcal{M} \to \mathbb{R}$ and $w : S \times \mathcal{M} \to \mathbb{R}$ and an aggregate law of motion $H : S \times \mathcal{M} \times S \to \mathcal{M}$ s.t.

(1) v, a', c are measurable with respect to $\mathcal{B}(S)$, v satisfies the household's Bellman equation and a', c are the associated policy functions, given r, w.

(2) K, L satisfy

$$r(s, \Phi) = F_K(K(s, \Phi), L(s, \Phi)) - \delta$$

$$w(s, \Phi) = F_L(K(s, \Phi), L(s, \Phi))$$

Definition of RCE (cont.)

(3) For all $\Phi \in \mathcal{M}$ and all $s \in S$

$$K'(s, \Phi) = \int a'(a, y, s, \Phi) d\Phi, \quad L(s, \Phi) = \int y d\Phi$$

$$\int c(a, y, s, \Phi) d\Phi + \int a'(a, y, s, \Phi) d\Phi = F(K(s, \Phi), L(s, \Phi)) + (1 - \delta) K(s, \Phi)$$

(4) The aggregate law of motion H is generated by the exogenous Markov process π and the policy function a'.

Transition Function and Law of Motion

- Define $Q_{\Phi_{s,s'}} : Z \times \mathcal{B}(Z) \to [0,1]$ by

$$Q_{\Phi_{s,s'}}((a,y),(\mathcal{A},\mathcal{Y})) = \sum_{y' \in \mathcal{Y}} \begin{cases} \pi(y', s' | y, s) & \text{if } a'(a,y,s,\Phi) \in \mathcal{A} \\ 0 & \text{otherwise} \end{cases}$$

- Aggregate law of motion

$$\Phi'(\mathcal{A},\mathcal{Y}) = \int Q_{\Phi_{s,s'}}((a,y),(\mathcal{A},\mathcal{Y}))\Phi(da \times dy)$$

Computation of the Recursive Equilibrium I

- Key computational problem: aggregate wealth distribution Φ is an infinite-dimensional object.
- Agents need to keep track of the aggregate wealth distribution to forecast future capital stock and thus future prices. But for K' we need entire Φ since
$$K' = \int a'(a, y, s, \Phi) d\Phi$$
- If a' were linear in a, exact aggregation would occur and K would be a sufficient statistic for K'.
- Idea: Approximate Φ with a finite set of moments.

Computation of the Recursive Equilibrium II

- Let the $n-$dimensional vector m denote the first n moments of the asset distribution.

- Agents use an approximate law of motion $m' = H_n(s, m)$.
- We choose the number of moments and the functional form of the function H_n.

- Krusell and Smith (1998) pick $n = 1$ and pose

$$\log(K') = a_s + b_s \log(K)$$

for $s \in \{s_b, s_g\}$. Here (a_s, b_s) are parameters.

Computation of the Recursive Equilibrium III

- Household problem

$$v(a, y, s, K) = \max_{c, a' \geq 0} \{U(c) + \beta \sum_{y' \in Y} \sum_{s' \in S} \pi(y', s' | y, s) v(a', y', s', K')\}$$

$$s.t.\ c + a' = w(s, K)y + (1 + r(s, K))a$$

$$\log(K') = a_s + b_s \log(K)$$

- Reduction of the state space to a four dimensional space $(a, y, s, K) \in \mathbb{R} \times Y \times S \times \mathbb{R}$.

Algorithm I

(1) Guess (a_s, b_s).

(2) Solve household problem to obtain $a'(a, y, s, K)$.

(3) Simulate economy for large number of T periods for large number N households.

- Start with initial conditions for the economy (s_0, K_0) and for each household (a_0^i, y_0^i).
- Draw random sequences $\{s_t\}_{t=1}^{T}$ and $\{y_t^i\}_{t=1,i=1}^{T,N}$ and use $a'(a, y, s, K)$ and perceived law of motion for K to generate sequences of $\{a_t^i\}_{t=1,i=1}^{T,N}$.
- Obtain $K_t = \frac{1}{N}\sum_{i=1}^{N} a_t^i$.

Algorithm II

(4) Run the regressions

$$\log(K') = \alpha_s + \beta_s \log(K)$$

to estimate (α_s, β_s) for $s \in S$.

(5) If the R^2 for this regression is high and $(\alpha_s, \beta_s) \approx (a_s, b_s)$, stop. An approximate equilibrium was found.

(6) Otherwise, update guess for (a_s, b_s). If guesses for (a_s, b_s) converge, but R^2 remains low, add higher moments to the aggregate law of motion and/or experiment with a different functional form.

Calibration I

- Period 1 quarter.
- CRRA utility with $\sigma = 1$ (log-utility)
- $\beta = 0.96$.
- $\alpha = 0.36$
- Aggregate component: two states (recession, expansion)

$$S = \{0.99, 1.01\} \Rightarrow \sigma_s = 0.01$$

- Symmetric transition matrix $\pi(s_g|s_g) = \pi(s_b|s_b)$.
- Expected time in each state: 8 quarters, hence $\pi(s_g|s_g) = \frac{7}{8}$.

Calibration II

- Idiosyncratic component: two states (employed/unemployed):

$$Y = \{0.25, 1\}$$

- Transition probabilities: $\pi(y', s'|y, s) = \pi(y'|s', y, s) \times \pi(s'|s)$.

- Expansion: average time of unemployment 1.5 quarters

$$\pi(y' = y_u|s' = s_g, y = y_u, s = s_g) = \frac{1}{3}$$
$$\pi(y' = y_e|s' = s_g, y = y_u, s = s_g) = \frac{2}{3}$$

- Recession: average time of unemployment 2.5 quarters

$$\pi(y' = y_u|s' = s_b, y = y_u, s = s_b) = 0.6$$
$$\pi(y' = y_e|s' = s_b, y = y_u, s = s_b) = 0.4$$

Calibration III

- From g to b: probability of remaining unemployed 1.25 higher

$$\pi(y' = y_u | s' = s_b, y = y_u, s = s_g) = 0.75$$
$$\pi(y' = y_e | s' = s_b, y = y_u, s = s_g) = 0.25$$

- From b to g: probability of remaining unemployed 0.75 higher

$$\pi(y' = y_u | s' = s_g, y = y_u, s = s_b) = 0.25$$
$$\pi(y' = y_e | s' = s_g, y = y_u, s = s_b) = 0.75$$

- Based on these, we can compute $\pi(y', s' | y, s)$.

Numerical Results

- The model delivers

(1) Aggregate law of motion

$$m' = H_n(s, m)$$

(2) Individual decision rules

$$a'(a, y, s, m)$$

(3) Time-varying cross-sectional wealth distributions $\Phi(a, y)$

Aggregate Law of Motion I

- Agents are bounded-rational.
- Suppose the converged law of motion:

$$\log(K') = 0.095 + 0.962 \log(K) \quad \text{for } s = s_g$$
$$\log(K') = 0.085 + 0.965 \log(K) \quad \text{for } s = s_b$$

- How irrational are agents? Use simulated time series $\{s_t, K_t\}_{t=0}^{T}$, divide sample into periods with $s_t = s_b$ and $s_t = s_g$, and run

$$\log(K_{t+1}) = \alpha_j + \beta_j \log(K_t) + \epsilon_{t+1}^{j}$$

Aggregate Law of Motion II

- Define

$$\widehat{\epsilon}^{j}_{t+1} = \log(K_{t+1}) - \widehat{\alpha}_j - \widehat{\beta}_j \log(K_t) \quad \text{for } j = g, b$$

- Then:

$$\sigma_j = \left(\frac{1}{T_j} \sum (\widehat{\epsilon}^{j}_{t})^2 \right)^{0.5}$$

$$R_j^2 = 1 - \frac{\sum (\widehat{\epsilon}^{j}_{t})^2}{\sum (\log K_{t+1} - \log \bar{K})^2}$$

- If $\sigma_j = 0$ for $j = g, b$ (if $R_j^2 = 1$ for $j = g, b$), then agents do not make forecasting errors.

Why Quasi-Aggregation?

- If all agents have linear savings functions with same marginal propensity to save

$$a'(a, y, s, K) = a_s + b_s a + c_s y$$

- Then:

$$K' = a_s + b_s \int a d\Phi + c_s \bar{L} = \widetilde{a}_s + b_s K$$

- Exact aggregation: K sufficient statistic for Φ.
- In this economy: savings functions almost linear with similar slope for $y = y_u$ and $y = y_e$.
- Only exceptions are unlucky agents $(y = y_u)$ with little assets. But these agents hold a negligible fraction of aggregate wealth and do not matter for K dynamics.

Illustration

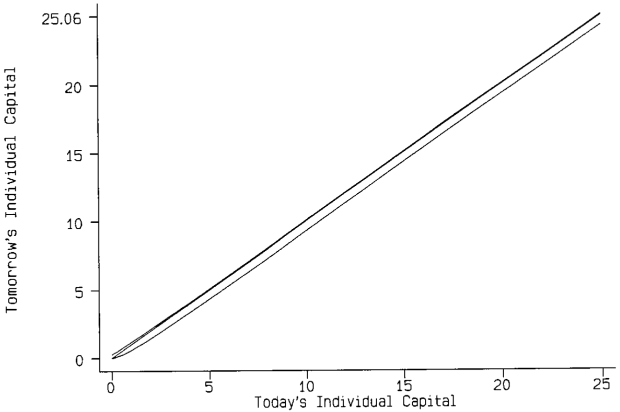


FIG. 2.—An individual agent's decision rules (benchmark model, aggregate capital = 11.7, good aggregate state).

How Important is Heterogeneity for Business Cycles?

- The experiment: compare two identical economies, except that one has complete markets (therefore no heterogeneity).
- Finding: Heterogeneity has little effect on the model's business cycle properties.

Illustration

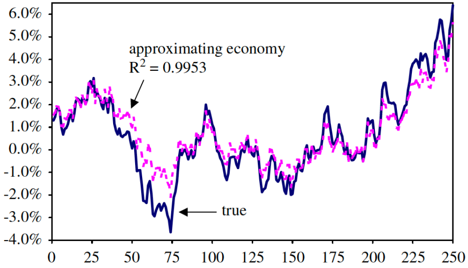
AGGREGATE TIME SERIES

Model	Mean(k_t)	Corr(c_t, y_t)	Standard Deviation (i_t)	Corr(y_t, y_{t-4})
Benchmark:				
Complete markets	11.54	.691	.031	.486
Incomplete markets	11.61	.701	.030	.481
$\sigma = 5$:				
Complete markets	11.55	.725	.034	.551
Incomplete markets	12.32	.741	.033	.524

Measure of the Accuracy of Aggregate Law of Motion

- Den Hann (2007) shows that solutions with high R^2 values may still be inaccurate according to more appropriate accuracy measures.

- R^2 is a measure in "average." Especially for this type of accuracy test, it is important to use stringent standards.



Measure of the Accuracy of Aggregate Law of Motion

- There is scaling issue with R^2. If we consider a regression with $K_{t+1} - K_t$ on K_t, R^2 can be substantially different from the level specification with K_{t+1} only.
- There is overfitting issue as R^2 can only increase as more explanatory variables are added.

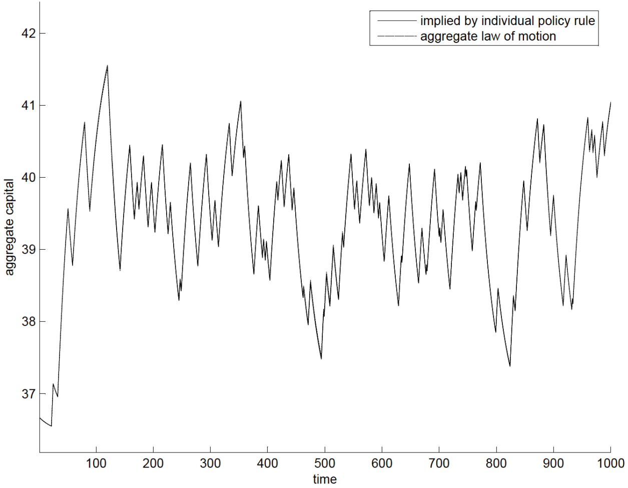
Measure of the Accuracy of Aggregate Law of Motion

- Instead, he proposes a powerful accuracy test which compares two aggregate capital series.

  (1) a series obtained by simulating a panel of agents using the individual policy rules.

  (2) a series obtained by the aggregate law of motion.

Figure 1. Accuracy of the aggregate law of motion under Algorithm 1 (random shocks).



Preference Heterogeneity

- The question:
  - The baseline model has far too little wealth heterogeneity.
  - Does approximate aggregate still hold when there is a realistic amount of wealth heterogeneity?
- The approach:
  - Add enough preference heterogeneity to the model to roughly replicate the observed distribution of wealth.
  - Check whether the mean is still enough to forecast prices very accurately.

Preference Heterogeneity

- Allow for 3 arbitrary value of β: 0.986, 0.989, 0.993.
- Agents switch β values stochastically, on average every 50 years (once per generation).
- Approximate aggregation is still very good having $R^2 \approx 0.999$.

Preference Heterogeneity

DISTRIBUTION OF WEALTH: MODELS AND DATA

Model	Percentage of Wealth Held by Top 1%	5%	10%	20%	30%	Fraction with Wealth < 0	Gini Coefficient
Benchmark model	3	11	19	35	46	0	.25
Stochastic-β model	24	55	73	88	92	11	.82
Data	30	51	64	79	88	11	.79

- The model with preference heterogeneity matches better the wealth distribution statistics from the data.
- Whether preference heterogeneity is important in the data is an open question.

What if Quasi-Aggregation fails?

- Different methods to approximate distributions
  - Moments: Krusell and Smith (1998), Algan, Allais, and Den Hann (2008), Winberry (2018)
  - Discretization: Reiter (2010)
  - Basis vectors/functions: Ahn et al. (2018), Childers (2018), Chang, Chen, and Schorfheide (2022)
  - Sequence-space Jacobian: Auclert et al. (2021)

Winberry (2018, QE)

- Finite-dimensional approximation of distributions:

$$p_{t,\epsilon}^{(K)}(x) = \exp\left\{\gamma_{t,\epsilon,0} + \gamma_{t,\epsilon,1}(x - m_{t,\epsilon,1}) \right.$$
$$\left. + \sum_{k=2}^{K} \gamma_{t,\epsilon,k}\left[(x - m_{t,\epsilon,1})^k - m_{t,\epsilon,k}\right]\right\}.$$

- $m_{t,\epsilon,k}$ are centralized moments of the distribution.
- Parameters $\gamma_{t,\epsilon,k}$ and moments $m_{t,\epsilon,k}$ must be consistent.

$$m_{t,\epsilon,1} = \int x p_{t,\epsilon}^{(K)}(x) dx, \quad m_{t,\epsilon,k} = \int (x - m_{t,\epsilon,1})^k p_{t,\epsilon}^{(K)}(x) dx.$$

- Density needs to integrate to one.

Unexpected Aggregate Shocks and Transition Dynamics

- Hypothetical thought experiment:
  - Economy is in stationary equilibrium, with a given policy.
  - Unexpectedly government policy changes. Exogenous change may be either tansitory or permanent.
  - Want to compute transition path induced by the change.
- Example: permanent introduction of a capital income tax at rate τ. Receipts are rebated to households as lump-sum transfers.
- Key: assume that after T periods the transition from old to new stationary equilibrium is completed.

Algorithm I

(1) Fix T.

(2) Compute stationary equilibrium $\Phi_0, v_0, r_0, w_0, K_0$ associated with $\tau = \tau_0 = 0$.

(3) Compute stationary equilibrium $\Phi_\infty, v_\infty, r_\infty, w_\infty, K_\infty$ associated with $\tau_\infty = \tau$. Assume:

$$\Phi_T, v_T, r_T, w_T, K_T = \Phi_\infty, v_\infty, r_\infty, w_\infty, K_\infty$$

(4) Guess sequence $\{\widehat{K}_t\}_{t=1}^{T-1}$. Note that $\widehat{K}_1$ is determined by decisions at time 0 and $L_t = L_0 = \bar{L}$ is fixed. Then:

$$\widehat{w}_t = F_L(\widehat{K}_t, \bar{L})$$
$$\widehat{r}_t = F_K(\widehat{K}_t, \bar{L}) - \delta$$
$$\widehat{T}_t = \tau_t \widehat{r}_t \widehat{K}_t$$

Algorithm II

(5) Since we know $v_T(a, y)$ and $\{\widehat{r}_t, \widehat{w}_t, \widehat{T}_t\}_{t=1}^{T-1}$, we can solve for $\{\widehat{v}_t, \widehat{c}_t, \widehat{a}_{t+1}\}_{t=1}^{T-1}$ backwards.

(6) With $\{\widehat{a}_{t+1}\}_{t=1}^{T-1}$ define transition laws $\{\widehat{\Gamma}_t\}_{t=1}^{T-1}$. Given the initial stationary equilibrium, we iterate forward from Φ_0 with

$$\widehat{\Phi}_{t+1} = \widehat{\Gamma}_t(\widehat{\Phi}_t) \quad \text{for } t = 1, \ldots, T-1.$$

(7) With $\{\widehat{\Phi}_t\}_{t=1}^{T}$, compute $\widehat{A}_t = \int a d\widehat{\Phi}_t$ for $t = 1, \ldots, T$.

(8) Check whether $\max_{1 \le t < T} |\widehat{A}_t - \widehat{K}_t| < \epsilon$. If yes, go to (9). If not, adjust your guesses for $\{\widehat{K}_t\}_{t=1}^{T-1}$ in (4).

(9) Check whether $|\widehat{A}_T - K_T| < \epsilon$. If yes, we are done. If not, go to (1) and increase T.

Welfare Consequences of the Policy Reform I

- Previous procedure determines aggregate variables such as r_t, w_t, Φ_t, K_t decision rules c_t, a_{t+1}, and value functions v_t.
- We can use v_0, v_1, and v_T to determine the welfare consequences from the reform.
- Suppose that $U(c) = \frac{c^{1-\sigma}}{1-\sigma}$.
- Optimal consumption allocation in initial stationary equilibrium, in sequential formulation, $\{c_s\}_{s=0}^{\infty}$.

$$v_0(a, y) = \mathbb{E}_0 \sum_{s=0}^{\infty} \beta^s \frac{c_s^{1-\sigma}}{1-\sigma}$$

Welfare Consequences of the Policy Reform II

- Define g implicitly as:

$$v_1(a,y) = v_0(a,y;g) = (1+g)^{1-\sigma}\mathbb{E}_0\beta^s \frac{c_s^{1-\sigma}}{1-\sigma}$$
$$= (1+g)^{1-\sigma} v_0(a,y)$$

- Then:

$$g(a,y) = \left[\frac{v_1(a,y)}{v_0(a,y)}\right]^{\frac{1}{1-\sigma}} - 1$$

- Steady state welfare consequences:

$$g_{ss}(a,y) = \left[\frac{v_T(a,y)}{v_0(a,y)}\right]^{\frac{1}{1-\sigma}} - 1$$

- $g(a,y)$ and $g_{ss}(a,y)$ may be quite different. example: social security reform.

Rules and Guidelines for Mini-Conference

- Both team members present.
- 20-minute presentation (2-min. & 1-min. warning, no interruption) + 5-minute discussion and Q&A.
- No more than 20 slides. Make sure that you have one-page conclusion slide to highlight the takeaways from the paper. (If you haven't gone through all the slides when you get 1-min. warning, you should go to the conclusion slide.)
- Key methodological/computational contribution of a paper should be presented.
- Be selective! You cannot present every detail in 20 minutes.